

Homework 5

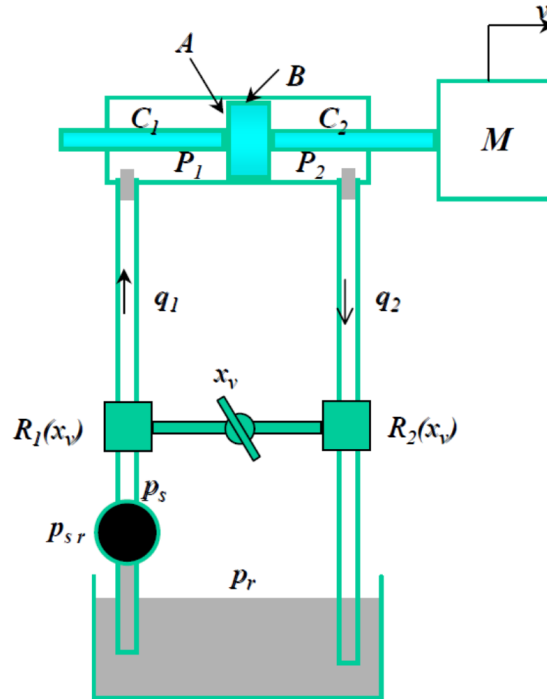
Jizhou Yan ^{TA}

Instructions before the assignment 作业前须知:

- This assignment is for the **sixth week's homework** of the System Dynamics and Control course (hereinafter referred to as SDC). The submission deadline is **before 11:59 PM on October 24, 2024**. Please ensure timely submission, as late submissions will result in point deductions. 本次作业为 SDC 第六周课程作业, 提交时间为 **2024/10/24 23:59** 前。
 - If submitted within 12 hours after the deadline, no points will be deducted (**this is allowed only twice, so focus on completing it properly rather than rushing to meet the deadline if you're unable to finish on time**);
 - If submitted within three days after the deadline, **20% of the total score of this assignment will be deducted**;
 - If submitted within two week after the deadline, **30% of the total score of this assignment will be deducted**;
 - Any other late submissions will result in a **40% deduction**;
- For late submissions, please contact the teaching assistant(TA), **Yan Jizhou**, via email at **yanjizhou312@163.com** or WeChat **y3139838452**. Make sure to use the filename format as "SDC-Assignment-x-Resubmission-Name-Student ID." 作业补交时通过邮箱或微信与助教闫济洲联系, 邮件标题及文件名备注为 "SDC-第 x 次作业补交-姓名-学号".
- The assignment is graded out of 10 points. It consists of basic course questions (10 points) and extension questions (0 to 5 points). **The final assignment score will be the smaller one between 10 and the actual score.** Each assignment earns all points by completing only the basic assignment. 每次作业仅完成基础作业即可获得所有分数, 鼓励同学们完成拓展课程题目。
- Students are encouraged to attempt the extension questions. Any extra points earned will be proportionally added to the final course grade for possible grade improvement.
- **Academic misconduct such as plagiarism is strictly prohibited in course work. Plagiarism and other academic misconduct will be dealt with in accordance with the relevant regulations of the University.** If you have discussed the assignment with classmates or received help, please note at the beginning of the assignment: "This assignment was completed after discussion with XXX (classmate's name), and the thought process was referenced from XXX (classmate's name)." **Every contribution deserves recognition. Unacknowledged similarities between assignments will be treated as plagiarism.** 课程作业中严禁抄袭等学术不端行为。如作业完成过程中经历了同学讨论/同学帮助的情况, 请在作业最前面注明 "本次作业经过与 XXX (同学姓名) 讨论, 参考了 XXX (同学姓名) 的部分思路"。每位同学的帮助都值得被记录。不被注明的作业雷同将被认为是抄袭行为。

1. Problem 1

Consider an inertia load M driven by a double-rod hydraulic cylinder as shown in the following figure. The fluid inside the hydraulic cylinder has an area of A (the ram area of the rod). The cylinder has a fluid capacitance of C_1 for its forward chamber and a fluid capacitance of C_2 for its return chamber. The damping friction coefficient of the rod is B . A pump is used to supply an input pressure p_s , which causes a forward flow q_1 into the hydraulic cylinder through a valve having a resistance of R_1 and a return flow of q_2 from the return chamber of the cylinder to the tank through a valve having a resistance of R_2 . The goal is to determine the supplied pressure $p_{sr}(t)$ such that the inertia load M can be positioned at any desired position as we want. To accomplish this objective, we need to model the system first. The input is p_{sr} and the output is the displacement of the inertia load x .



(A) Write down the elemental equations as well as interconnecting equations for the hydraulic subsystem, the cylinder, and the mechanical subsystem respectively.

(B) Transform the equations in (A) into algebraic equations in s-domain by using Laplace transform. Based on the transformed equations, draw the block diagram of the system. Indicate the hydraulic subsystem, the cylinder, and the mechanical subsystem on the block diagram.

(C) Obtain the transfer function from the input to the output.

(D) Is the system stable? Can the system stay at a particular position by applying a constant supplied pressure p_{sr} ?

2. Problem 2

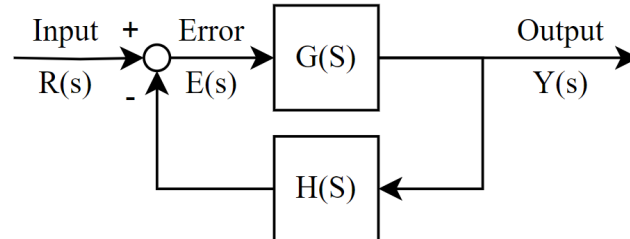
In class, We learned about basic closed-loop and open-loop control systems. Now, let's engage in some simple explorations.

2.1 a

A simple open-loop control system is written as



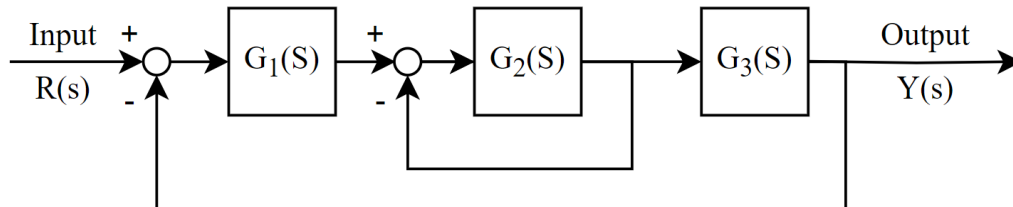
In many cases, this open-loop system may be unstable. By adding a feedback process to form a closed-loop control system, the system can become more stable.



Given $G(s) = 2s + 3$ and $H(s) = Ks$, discuss the conditions under which the closed-loop system is stable. If the system is required to have an overshoot less than 10% please sketch the trajectory of the poles as K varies.

2.2 b Cascade control

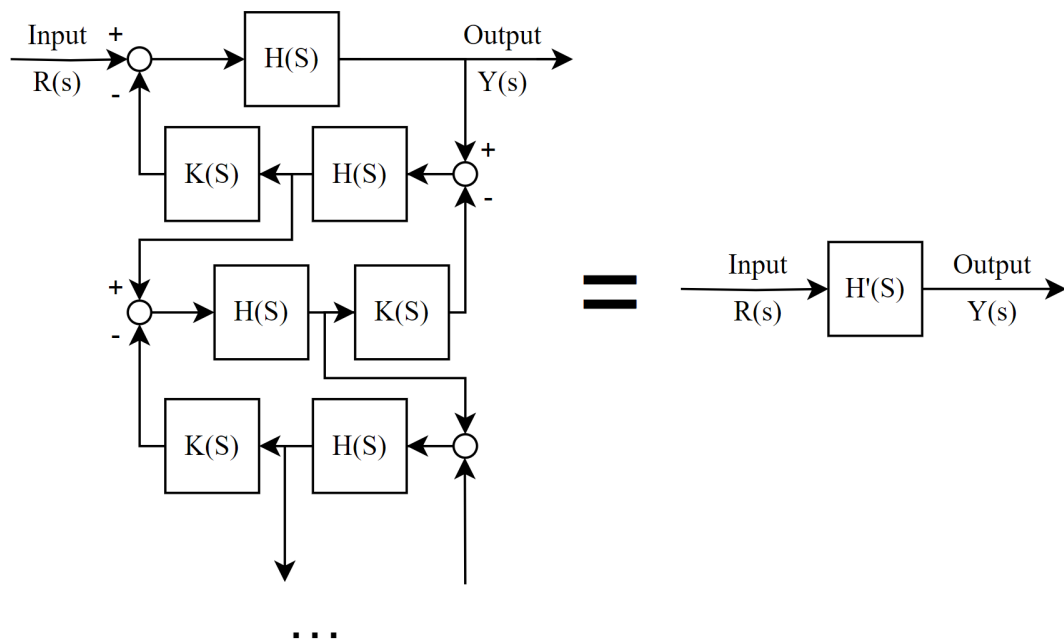
Let's return to our closed-loop system with feedback control. Some classic controllers, such as the cascaded PID controller, contain more than one closed loop. Please derive the transfer function of the cascaded controller shown in the diagram below.



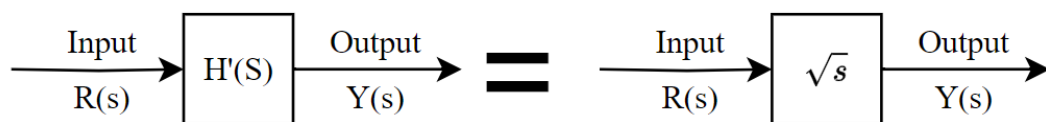
2.3 c obtain irrational transfer functions using rational transfer functions.

Generally, we usually use rational functions as transfer functions because they directly represent differentiation or integration of the input (Extension: think about what $\frac{1}{s+1}$ represents?). Rational transfer functions can be directly constructed through circuits. However, we can still try to construct irrational functions using rational functions.

By constructing the following infinitely large block diagram, can you equivalently write it as a simpler feedback loop?



Furthermore, can you adjust the relationship between $K(s)$ and $H(s)$ so that the equivalent transfer function becomes $H' = \sqrt{s}$?



$\sqrt{s}$ represents taking a fractional-order derivative of the input, which is an extension of differential operators. It is often used in circuit transmission analysis or in the field of quantum mechanics. An example is $\left(\frac{d}{dx}\right)^{\frac{1}{2}} x = \frac{2\sqrt{x}}{\sqrt{\pi}}$.

3. *Problem 3 (Extension)

Use the two-port network method mentioned in the previous assignment to solve the first question of this assignment